

Please write your names in CAPITAL letters.

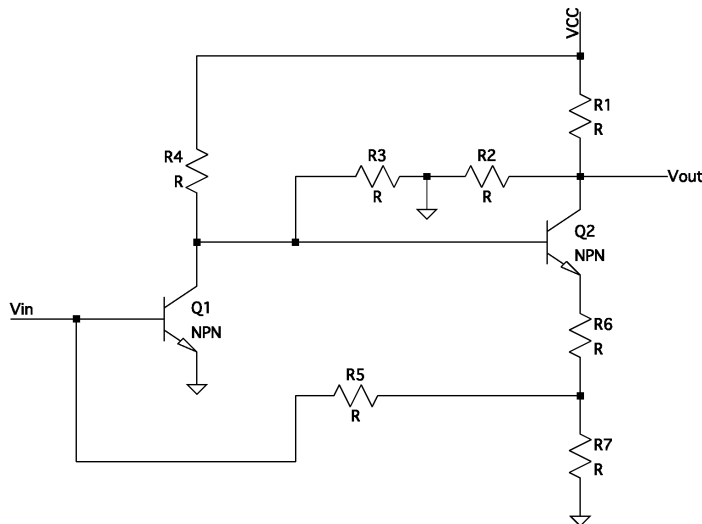
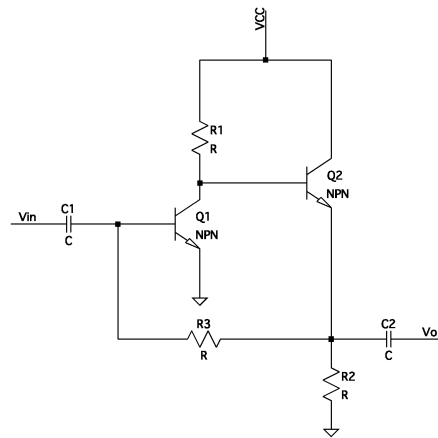
LAST NAME : ANSWER KEY
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SET A

Multiple Choice. Shade the correct answer/s in your answer sheet. Each item is worth 2 points.

- If an amplifier has a forward gain of 400 and the feedback ratio is 0.1. Find the overall gain with negative feedback.
a. 6.75 b. 7.75 **c. 9.75** d. 10.75
- A shunt series feedback amplifier has a voltage gain with feedback of 83.33 and feedback factor of 0.01. What is the voltage gain of the amplifier without feedback?
a. 200 c. 400
b. 300 **d. 500**
- What is the feedback configuration of the circuit on the right?
a. Shunt-Series
b. Series-Series
c. Series-Shunt
d. Shunt-Shunt



- What feedback circuits are present in the circuit on the left?
a. shunt-shunt and shunt-series
b. shunt-shunt only
c. shunt-series only
d. shunt-shunt and series-series

- Consider a series-series feedback network. It has an amplifier gain of 100, feedback factor of 5. For the basic amplifier, the input voltage = 4 V and the input current = 2 mA. Find the input resistance of the network.
a. 1002 k Ω b. 2 k Ω c. 1.002 k Ω d. 2000 k Ω
- What happens to the impedances of an amplifier where the feedback network is connected in parallel with the output and in series with the input? Shade all answers that apply.
a. Its input impedance increases
b. Its output impedance increases
c. Its output impedance decreases
d. Its input impedance decreases

no partial points

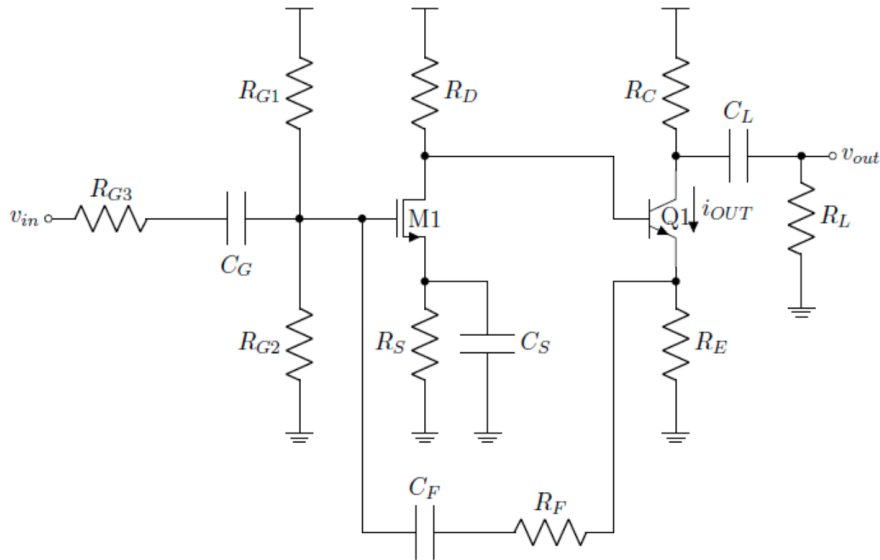
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7. The negative feedback in an amplifier leads to which one of the following?
- Increase in current gain
 - Increase in voltage gain
 - Decrease in voltage gain
 - Decreases in bandwidth

For items 8-12, consider the circuit below. Assume that the transistors are biased in their desired region of operation. Assume that capacitors can be treated as shorts during small-signal analysis.



$$\begin{array}{lll} R_C = 15 \text{ k}\Omega & R_{G1} = 5 \text{ k}\Omega & R_S = 100 \Omega \\ R_D = 83.3 \text{ k}\Omega & R_{G2} = 1 \text{ k}\Omega & g_{m,M1} = 625 \text{ mS} \\ R_E = 1 \text{ k}\Omega & R_{G3} = 2.5 \text{ k}\Omega & g_{m,Q1} = 40 \text{ mS} \\ R_F = 1.5 \text{ k}\Omega & R_L = 5 \text{ k}\Omega & \beta_{Q1} = 99 \end{array} \quad \begin{array}{l} V_{A,Q1} \rightarrow \infty \\ \lambda_{M1} = 0 \text{ V}^{-1} \end{array}$$

- What is the value of the open-loop input resistance?
 - 300 Ohms
 - 400 Ohms
 - 500 Ohms
 - 600 Ohms
- What is the value of the open loop transconductance?
 - 33.7 S
 - 35.7 S
 - 37.7 S
 - 39.7 S
- Determine the value of the feedback factor?
 - 0.1
 - 0.2
 - 0.3
 - 0.4
- What is the value of the open-loop gain?
 - 17,850
 - 17,850
 - 15,320
 - 15,320
- What is the value of the closed loop gain?
 - 1.3998
 - 1.3998
 - 2.4997
 - 2.4997
- Which of the following best describes the signal and feedback type in a shunt-shunt feedback amplifier?
 - Voltage input, voltage feedback
 - Current input, voltage feedback
 - Voltage input, current feedback
 - Current input, current feedback

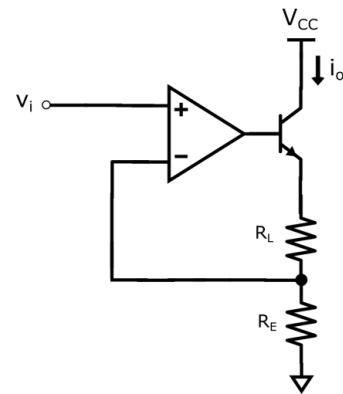
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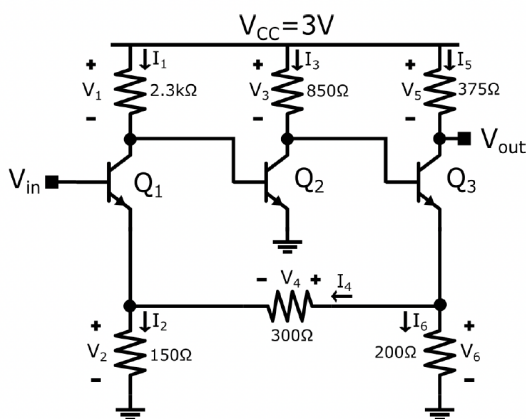
14. Identify the amplifier property that is applicable to a current amplifier.
- High input impedance
 - High efficiency
 - Low input impedance
 - Low output impedance
15. A shunt-series topology is a _____ amplifier with feedback.
- current
 - transconductance
 - voltage
 - transresistance
16. A transconductance amplifier should have:
- Low output impedance
 - High efficiency
 - Low input impedance
 - High output impedance

For items 17-18, please refer to the circuit on the right. The feedback amplifier has an $R_L = 1000$ ohms and $R_E = 2500$ ohms.



17. The circuit shown above is an example of what type of amplifier?
- Series – Series
 - Shunt – Series
 - Shunt – Shunt
 - Series – Shunt
18. Determine the feedback factor F for the circuit.
- 2500 ohms
 - 1000 ohms
 - 3500 ohms
 - 714 ohms

For items 19-22, refer the circuit below.



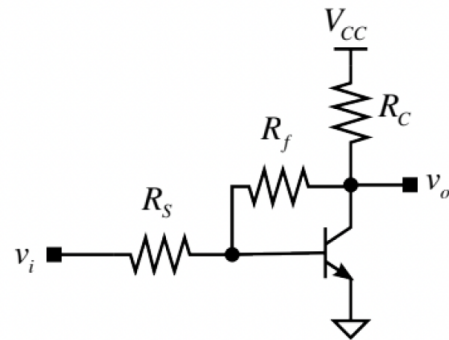
19. The circuit shown is an example of what type of amplifier?
- Series – Series
 - Shunt – Series
 - Shunt – Shunt
 - Series – Shunt
20. What is the input resistance of the feedback network in this circuit?
- 138.46 ohms
 - 500 ohms
 - 115.38 ohms
 - 650 ohms
21. What is the output resistance of the feedback network in this circuit?
- 138.46 ohms
 - 500 ohms
 - 115.38 ohms
 - 650 ohms
22. What is the feedback factor F in this circuit?
- 34.62
 - 80
 - 40
 - 46.15

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For items 23-26, consider the circuit on the right.



23. The circuit shown is an example of what type of amplifier?
- Series – Series
 - Shunt – Series
 - ☒ Shunt – Shunt
 - Series – Shunt
24. This amplifier is an example of a:
- current amplifier
 - transconductance amplifier
 - voltage amplifier
 - ☒ transresistance amplifier
25. If $R_s = 3000$ ohms and $R_f = 1000$ ohms, F is equal to:
- 1000
 - ☒ -0.001
 - 3000
 - ~~0.0003~~ 0.001
26. If $R_s = 3000$ ohms and $R_f = 1000$ ohms, the output resistance is:
- 4000 ohms
 - ☒ 1000 ohms
 - 3000 ohms
 - 0.001 ohms
27. Consider the following statements:
- A single-pole forward path will result in a stable loop gain for a passive feedback path implementation.
 - A feedback system with poles in the forward path will result in having only zeroes in the closed loop system.
 - A right-half plane pole in the loop gain will result in a stable oscillation at the system's output.
 - The dominant pole is the -3dB bandwidth for a single-pole system.

Which of the statement/s is/are always TRUE?

- I, II, III
- I, II, IV
- II, III
- ☒ I, IV

For items 28-31: An amplifier with a voltage gain of 20,000 V/V has three poles at 100 krad/s, 3 Mrad/s, and 5 Mrad/s. The amplifier is operated in the negative feedback configuration and with a feedback factor $F = 0.5$.

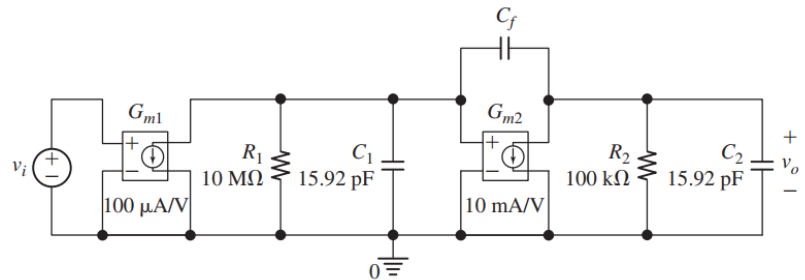
28. What is the unity-gain frequency ω_u of the loop gain?
- 31.07 Mrad/s
 - ☒ 24.66 Mrad/s
 - 19.57 Mrad/s
 - 39.15 Mrad/s
29. Calculate the phase margin of the loop-gain.
- ☒ -71.37°
 - 75.16°
 - -75.16°
 - 71.37°
30. Suppose that a narrowband technique is employed in the loop gain, calculate the new dominant pole if the target new unity gain frequency is $\omega_u = 1.4$ Mrad/s.
- 190 rad/s
 - 310 rad/s
 - 240 rad/s
 - ☒ 140 rad/s
31. Calculate the new phase margin of the loop gain after the narrowbanding compensation.
- 64.98°
 - 53.43°
 - ☒ 49.35°
 - 69.07°

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For items 32-34, refer to the two port of the two-stage amplifier figure on the right. Let $C_f = 2 \text{ pF}$.



32. What is the equivalent capacitance seen at the input of the second stage?
 - a. 15.92 pF
 - ☒ b. 2.02 nF
 - c. 2.00 nF
 - d. 17.92 pF
33. Determine the dominant pole of the Miller-compensated circuit?
 - ☒ a. 7.96 Hz
 - b. 6.28 kHz
 - c. 50 Hz
 - d. 999.72 Hz
34. Ignoring any zeroes of the amplifier, calculate the phase margin of the circuit.
 - a. 59.92°
 - b. 57.26°
 - ☒ c. 51.59°
 - d. 55.69°
35. In a Miller-compensated circuit, what is the trade-off when increasing the Miller capacitor (C_m)?
 - a. Higher power consumption vs. higher phase margin
 - b. Higher gain vs. lower bandwidth
 - ☒ c. Lower unity-gain frequency vs. higher stability
 - d. Lower phase margin vs. higher bandwidth
36. In a compensated two-pole system, what is the approximate phase margin if the second pole lies at 20x the unity-gain frequency?
 - ☒ a. 90°
 - b. 60°
 - c. 45°
 - d. 0°
37. Which of the following statements is true regarding dominant pole compensation?
 - a. Improves gain at high frequencies
 - ☒ b. Reduces the effect of higher frequency poles
 - c. Decreases the other poles in the system
 - d. Increases phase margin by adding LHP zeroes
38. An underdamped two-pole system has:
 - a. Real negative poles
 - b. Real and complex poles
 - c. Real positive poles
 - ☒ d. Complex conjugate poles
39. In a feedback closed-loop system, stability problems arises when:
 - ☒ a. Feedback becomes positive at high frequencies
 - b. Phase margin exceeds 90°
 - c. Loop gain is too small
 - d. Closed-loop gain bandwidth becomes too large
40. A feedback amplifier has an open-loop DC gain of 100 dB and poles at 1 kHz and 1 MHz. What is the gain-bandwidth product (GBW)?
 - a. 100 kHz
 - b. 1 MHz
 - c. 10 MHz
 - ☒ d. 100 MHz

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41. A feedback amplifier has an open-loop transfer function with poles at 10 kHz and 1 M Hz. If the phase shift at 1 MHz is 135°, what is the phase margin?
a. 0° **b. 45°** c. 90° d. 180°
42. If the closed-loop bandwidth of a unity-gain feedback amplifier is 100 kHz and the DC open-loop gain is 100, what is the open-loop gain-bandwidth product?
a. 100 Hz b. 1 kHz **c. 100 kHz** d. 1 MHz
43. The transfer function of an uncompensated amplifier is $A(s) = \frac{10^4}{(1+s/10^5)(1+s/10^6)}$. To improve stability, a capacitor is added using Miller compensation. What effect does this have?
a. Reduces DC gain
b. Shifts dominant pole lower
c. Increases unity-gain bandwidth
d. Decreases phase margin
44. Given a two-pole system with poles at 100 Hz and 1 kHz, what compensation technique can be used to ensure stability?
a. Increase gain c. Add zero at 1 MHz
b. Miller compensation d. Decrease DC gain
45. A feedback amplifier has a step response with 25% overshoot. Estimate the phase margin.
a. 30° **b. 45°** c. 60° d. 75°
46. What is the primary reason for using feedback in amplifiers with respect to frequency response?
a. Increase gain **c. Improve linearity and stability**
b. Reduce input impedance d. Decrease bandwidth
47. A low phase margin usually indicates:
a. A more stable system **c. Potential for oscillation and ringing**
b. A faster system with no overshoot d. Better low-frequency response
48. Narrowbanding compensation is most effective for:
a. Increasing bandwidth **c. Isolating high-frequency poles**
b. Reducing loop gain d. Preventing low-frequency oscillations
49. Which of the following best describes Miller compensation?
a. It increases the output resistance **c. It creates a dominant low-frequency pole**
b. It adds a zero to the loop gain d. It decreases power consumption
50. What happens to the step response of a system as phase margin increases?
a. Overshoot increases **c. Overshoot decreases**
b. Rise time decreases d. System becomes underdamped

B or C
gets full
points

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51. Why is phase margin important in feedback amplifiers?

- a. It determines power efficiency
- b. It directly affects the bandwidth
- c. It controls the steady-state error
- ☒ d. It predicts stability and transient response

52. Which of the following statements correctly distinguishes between narrowbanding and Miller compensation techniques used in feedback amplifiers?

- a. Narrowbanding is used to increase the bandwidth of amplifiers, while Miller compensation is used to reduce high-frequency gain.
- ☒ b. Narrowbanding involves reducing the bandwidth of a system to enhance selectivity, whereas Miller compensation introduces a dominant pole to stabilize multi-stage amplifiers.
- c. Narrowbanding adds a compensating capacitor between the input and output of an amplifier, while Miller compensation adjusts the biasing to reduce noise.
- d. Both narrowbanding and Miller compensation are exclusively used for RF amplifier design to improve impedance matching.

----- end of EEE 141 Exam 3 -----